$$\begin{array}{c|cc} & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-\,\alpha\beta\chi} \\ \hline \mathcal{K}^{\#1}_{0^+}{}^{\dagger} & -\frac{Y_1 k^2}{2} & 0 \\ \mathcal{K}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0 \end{array}$$

$$\begin{array}{c|cccc} & \mathcal{K}^{\#1}_{1^+\,\alpha\beta} & \mathcal{K}^{\#2}_{1^+\,\alpha\beta} & \mathcal{K}^{\#1}_{1^-\,\alpha} & \mathcal{K}^{\#2}_{1^-\,\alpha} \\ \hline \mathcal{K}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & -\frac{Y_2 k^2}{2} & \frac{k^2 \overset{(4)}{\kappa}_{4}}{2\sqrt{2}} & 0 & 0 \\ \mathcal{K}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{k^2 \overset{(4)}{\kappa}_{4}}{2\sqrt{2}} & \frac{Y_3 k^2}{2} & 0 & 0 \\ \mathcal{K}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & 0 & 0 \\ \mathcal{K}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & 0 & 0 \end{array}$$

$$\begin{array}{c|cc} & \mathcal{K}^{\#1}_{2^+\,\alpha\beta} & \mathcal{K}^{\#1}_{2^-\,\alpha\beta\chi} \\ \hline \mathcal{K}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & -\frac{Y_4 k^2}{2} & 0 \\ \mathcal{K}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0 \end{array}$$

$$\begin{array}{c|cc} & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-\,\alpha\beta\chi} \\ \hline \mathcal{J}^{\#1}_{0^+}{}^{\dagger} & \frac{1}{\mathrm{Det}(0^+)} & 0 \\ \mathcal{J}^{\#1}_{0^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0 \end{array}$$

$$\begin{array}{c|cccc} & \mathcal{J}^{\#1}_{1^+\,\alpha\beta} & \mathcal{J}^{\#2}_{1^+\,\alpha\beta} & \mathcal{J}^{\#1}_{1^-\,\alpha} & \mathcal{J}^{\#2}_{1^-\,\alpha} \\ \hline \mathcal{J}^{\#1}_{1^+}{}^{\dagger\,\alpha\beta} & \frac{Y_3 k^2}{2\,\mathrm{Det}(1^+)} & -\frac{k^2 \overset{(4)}{\kappa}_{4}}{2\sqrt{2}\,\mathrm{Det}(1^+)} & 0 & 0 \\ \mathcal{J}^{\#2}_{1^+}{}^{\dagger\,\alpha\beta} & -\frac{k^2 \overset{(4)}{\kappa}_{4}}{2\sqrt{2}\,\mathrm{Det}(1^+)} & -\frac{Y_2 k^2}{2\,\mathrm{Det}(1^+)} & 0 & 0 \\ \mathcal{J}^{\#1}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & 0 & 0 \\ \mathcal{J}^{\#2}_{1^-}{}^{\dagger\,\alpha} & 0 & 0 & 0 & 0 \end{array}$$

$$\begin{array}{c|cc} & \mathcal{J}^{\#1}_{2^+\,\alpha\beta} & \mathcal{J}^{\#1}_{2^-\,\alpha\beta\chi} \\ \hline \mathcal{J}^{\#1}_{2^+}{}^{\dagger\,\alpha\beta} & \frac{1}{\mathrm{Det}(2^+)} & 0 \\ \mathcal{J}^{\#1}_{2^-}{}^{\dagger\,\alpha\beta\chi} & 0 & 0 \end{array}$$

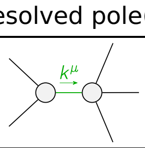
Abbreviations used in matrices

$$Y_1 == 3\,\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{6}\ \&\&\ Y_2 == \overset{(4)}{\kappa}_{3} - \overset{(4)}{\kappa}_{6}\ \&\&$$
$$Y_3 == \overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4}\ \&\&\ Y_4 == \overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{6}\ \&\&\ \mathrm{Det}(0^+) == -\frac{1}{2} k^2 (3\,\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{6})\ \&\&$$
$$\mathrm{Det}(1^+) == -\frac{1}{8} k^4 \left(2\,\overset{(4)}{\kappa}_{3}{}^2 + 2\,\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{4} + \overset{(4)}{\kappa}_{4}{}^2 - 2(\overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4})\overset{(4)}{\kappa}_{6}\right)\ \&\&\ \mathrm{Det}(2^+) == -\frac{1}{2} k^2 (\overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{6})$$

Lagrangian

$$\overset{(4)}{\kappa}_{1}\,\partial_\beta \mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_{3}\,\partial_\beta \mathcal{K}^{\alpha\beta\chi}\,\partial_\delta \mathcal{K}_{\alpha\chi}{}^{\delta} + \overset{(4)}{\kappa}_{4}\,\partial_\alpha \mathcal{K}^{\alpha\beta\chi}\,\partial_\delta \mathcal{K}_{\beta\chi}{}^{\delta} +$$
$$\overset{(4)}{\kappa}_{6}\,\partial_\beta \mathcal{K}^{\alpha\beta\chi}\,\partial_\delta \mathcal{K}_{\chi\alpha}{}^{\delta} + \frac{1}{2}\overset{(4)}{\kappa}_{3}\,\partial_\alpha \mathcal{K}^{\alpha\beta\chi}\,\partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} + \frac{1}{2}\overset{(4)}{\kappa}_{4}\,\partial_\alpha \mathcal{K}^{\alpha\beta\chi}\,\partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\ \mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	**# constraint(s)**	**Covariant form**
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{0^-} == 0$	1	$\partial_\delta\partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta \mathcal{J}^{\beta\alpha\chi} == \partial_\delta\partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta \mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}^{\#2\,\alpha}_{1^-} == 0$	3	$\partial_\chi\partial_\beta \mathcal{J}^{\beta\alpha\chi} == 0$
$\mathcal{J}^{\#1\,\alpha}_{1^-} == 0$	3	$\partial_\chi\partial^\alpha \mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi \mathcal{J}^{\beta\alpha}{}_{\beta} == \partial_\chi\partial_\beta \mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}^{\#1\,\alpha\beta\chi}_{2^-} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha \mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi \mathcal{J}^{\beta\alpha\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha \mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon \mathcal{J}^{\delta\alpha}{}_{\delta} == 3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta \mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi \mathcal{J}^{\alpha\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta \mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon \mathcal{J}^{\delta\beta}{}_{\delta}$
Total # constraint(s):	12	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$-\frac{4\,\overset{(4)}{\kappa}_{3}{}^2 + 4\,\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{4} + 3\,\overset{(4)}{\kappa}_{4}{}^2 - 8\,\overset{(4)}{\kappa}_{4}\overset{(4)}{\kappa}_{6} + 8\,\overset{(4)}{\kappa}_{6}{}^2}{(2\,\overset{(4)}{\kappa}_{3}{}^2 + \overset{(4)}{\kappa}_{4}(\overset{(4)}{\kappa}_{4} - 2\,\overset{(4)}{\kappa}_{6}) + 2\,\overset{(4)}{\kappa}_{3}(\overset{(4)}{\kappa}_{4} - \overset{(4)}{\kappa}_{6}))(\overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{6})}$

Resolved unitarity condition(s):

$$(\overset{(4)}{\kappa}_{4} < 0\ \&\&$$
$$((\overset{(4)}{\kappa}_{3} < -\frac{\overset{(4)}{\kappa}_{4}}{2}\ \&\&\ \frac{2\,\overset{(4)}{\kappa}_{3}{}^2 + 2\,\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{4} + \overset{(4)}{\kappa}_{4}{}^2}{2\,\overset{(4)}{\kappa}_{3} + 2\,\overset{(4)}{\kappa}_{4}} < \overset{(4)}{\kappa}_{6} < -\overset{(4)}{\kappa}_{3})\ ||\ (-\frac{\overset{(4)}{\kappa}_{4}}{2} < \overset{(4)}{\kappa}_{3} < -\overset{(4)}{\kappa}_{4}\ \&\&\ \frac{2\,\overset{(4)}{\kappa}_{3}{}^2 + 2\,\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{4} + \overset{(4)}{\kappa}_{4}{}^2}{2\,\overset{(4)}{\kappa}_{3} + 2\,\overset{(4)}{\kappa}_{4}} < \overset{(4)}{\kappa}_{6} < -\overset{(4)}{\kappa}_{3})\ ||$$
$$(\overset{(4)}{\kappa}_{3} == -\overset{(4)}{\kappa}_{4}\ \&\&\ \overset{(4)}{\kappa}_{6} < -\overset{(4)}{\kappa}_{3})\ ||\ (\overset{(4)}{\kappa}_{3} > -\overset{(4)}{\kappa}_{4}\ \&\&\ (\overset{(4)}{\kappa}_{6} < -\overset{(4)}{\kappa}_{3}\ ||\ \overset{(4)}{\kappa}_{6} > \frac{2\,\overset{(4)}{\kappa}_{3}{}^2 + 2\,\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{4} + \overset{(4)}{\kappa}_{4}{}^2}{2\,\overset{(4)}{\kappa}_{3} + 2\,\overset{(4)}{\kappa}_{4}})))\ ||$$
$$(\overset{(4)}{\kappa}_{4} == 0\ \&\&\ ((\overset{(4)}{\kappa}_{3} < 0\ \&\&\ \overset{(4)}{\kappa}_{3} < \overset{(4)}{\kappa}_{6} < -\overset{(4)}{\kappa}_{3})\ ||\ (\overset{(4)}{\kappa}_{3} > 0\ \&\&\ (\overset{(4)}{\kappa}_{6} < -\overset{(4)}{\kappa}_{3}\ ||\ \overset{(4)}{\kappa}_{6} > \overset{(4)}{\kappa}_{3}))))\ ||$$
$$(\overset{(4)}{\kappa}_{4} > 0\ \&\&\ ((\overset{(4)}{\kappa}_{3} < -\overset{(4)}{\kappa}_{4}\ \&\&\ \frac{2\,\overset{(4)}{\kappa}_{3}{}^2 + 2\,\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{4} + \overset{(4)}{\kappa}_{4}{}^2}{2\,\overset{(4)}{\kappa}_{3} + 2\,\overset{(4)}{\kappa}_{4}} < \overset{(4)}{\kappa}_{6} < -\overset{(4)}{\kappa}_{3})\ ||$$
$$(\overset{(4)}{\kappa}_{3} == -\overset{(4)}{\kappa}_{4}\ \&\&\ \overset{(4)}{\kappa}_{6} < -\overset{(4)}{\kappa}_{3})\ ||\ (-\overset{(4)}{\kappa}_{4} < \overset{(4)}{\kappa}_{3} < -\frac{\overset{(4)}{\kappa}_{4}}{2}\ \&\&\ (\overset{(4)}{\kappa}_{6} < -\overset{(4)}{\kappa}_{3}\ ||\ \overset{(4)}{\kappa}_{6} > \frac{2\,\overset{(4)}{\kappa}_{3}{}^2 + 2\,\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{4} + \overset{(4)}{\kappa}_{4}{}^2}{2\,\overset{(4)}{\kappa}_{3} + 2\,\overset{(4)}{\kappa}_{4}}))\ ||$$
$$(\overset{(4)}{\kappa}_{3} == -\frac{\overset{(4)}{\kappa}_{4}}{2}\ \&\&\ (\overset{(4)}{\kappa}_{6} < \frac{2\,\overset{(4)}{\kappa}_{3}{}^2 + 2\,\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{4} + \overset{(4)}{\kappa}_{4}{}^2}{2\,\overset{(4)}{\kappa}_{3} + 2\,\overset{(4)}{\kappa}_{4}}\ ||\ \overset{(4)}{\kappa}_{6} > \frac{2\,\overset{(4)}{\kappa}_{3}{}^2 + 2\,\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{4} + \overset{(4)}{\kappa}_{4}{}^2}{2\,\overset{(4)}{\kappa}_{3} + 2\,\overset{(4)}{\kappa}_{4}}))\ ||$$
$$(\overset{(4)}{\kappa}_{3} > -\frac{\overset{(4)}{\kappa}_{4}}{2}\ \&\&\ (\overset{(4)}{\kappa}_{6} < -\overset{(4)}{\kappa}_{3}\ ||\ \overset{(4)}{\kappa}_{6} > \frac{2\,\overset{(4)}{\kappa}_{3}{}^2 + 2\,\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{4} + \overset{(4)}{\kappa}_{4}{}^2}{2\,\overset{(4)}{\kappa}_{3} + 2\,\overset{(4)}{\kappa}_{4}}))))$$